SOCDet: A Lightweight and Accurate Oriented Object Detection Network for Satellite On-Orbit Computing

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***Abstract*— In recent years, the performance of the deep-learning-based object detection models for remote-sensing images (RSIs) improves with the increase of the hyperparameter scale. However, the state-of-the-art detection models are generally too cumbersome to adapt to the resource-constrained satellite platforms, and even general lightweight detection models cannot satisfy performance requirements. To solve those problems, a lightweight and accurately oriented object detection network for satellite on-orbit computing (SOCDet) is proposed from three levels. At the computing unit level, we propose an efficient computing unit Single-kernel Omni-dimensional Dynamic Convolution (SODC) to make SOCDet feature extraction more efficient. At the network module level, we design a structural reparameterization block (SRB) based on composite structure reparameterization to improve inference accuracy. A guided attention module (GAM) for guiding SOCDet is designed to extract object features. We design a lightweight and concise one-stage detection architecture at the network architecture level to accommodate satellite platforms with extremely constrained computing and storage resources. We conduct ablation experiments on the ground server and compare experiments with the state-of-the-art methods on embedded GPU, using DOTA, HRSC2016, and FAIR1M. The experimental results show that SOCDet can achieve 2.36× faster than the baseline in inference speed with 10.78 more mAP, 72.6% fewer Params, and 92.56% fewer floating-point operations (FLOPs). Compared with state-of-the-art methods, SOCDet improves the inference speed and derives competitive mAP results with affordable computing capability.**

***Index Terms*— Lightweight neural networks, oriented object detection, remote-sensing images (RSIs).**

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I. Introduction

SATELLITE on-orbit computing is promising to become the future satellite development trend, and satellite on-orbit object detection, an important part of satellite on-orbit computing, is also promising. As satellites are facing the issue of limited on-board processing abilities for resource-consuming artificial intelligence (AI) deployment, various schemes have been designed to solve this problem. NASA [1] investigates CubeSat-based edge Tensor Processing Unit (TPU) processors for autonomous operations and onboard science-data analysis from the perspective of satellite on-orbit computing equipment. ESA [2] has launched ϕ-Sat-1 which is the first onboard deep neural network demonstrator for satellite earth observation. The former studies high-performance onboard TPU to provide hardware support for AI deployment on satellites, while the latter gives an example of AI performing cloud detection on satellites. The study of leading space agencies in satellite deployment shows the necessity of developing deployable AI models on satellites. Our latest research, SOCNet [3], implements lightweight real-time object recognition for satellite on-orbit computing in resource-constrained situations, further demonstrating the feasibility of satellite on-orbit computing. Continuous improvement of the resolution of remote-sensing images (RSIs) is both an opportunity and a challenge for the research of RSI interpretation [4], [5]. On the one hand, high-resolution RSIs provide perfect data support for the object detection method based on deep learning, which is an opportunity for its development. On the other hand, high-resolution RSIs consume a large amount of data storage space and require high transmission bandwidth, which poses a challenge for the on-orbit resources of satellites. For example, the accurate data volume of high-resolution RSIs is several hundred Gigabit Per Second (Gb/s), but the storage on the satellite is only about hundreds of terabytes. Even if all the storage space of the satellite is used to store RSIs, the storage space of the satellite will be exhausted within tens of minutes. Meanwhile, the downlink transmission bandwidth is at most tens of Gb/s, which is also impossible to transmit all RSIs within the visible time. However, there is another challenge for the limited computation, satellite only has limited

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power and faces heat dissipation problems. Based on the above analysis, the cumbersome ground-based object detection models cannot properly operate on satellites. Satellite on-orbit computing is an effective way to solve the above bottlenecks, which provides accurate object detection on satellite, sends the results to the ground center, and requires no extra storage space and transmission bandwidth for users.

The foremost challenge of satellite on-orbit computing is that state-of-the-art detection models with massive parameters cannot adapt to satellite platforms with extremely limited resources. Generally speaking, object detection performance increases with the scale of hyperparameters, and various large detection models have been proposed. These large object detection methods for RSIs can be divided into two categories according to anchor boxes: horizontal bounding boxes (HBB) [6] and oriented bounding boxes (OBB) [7], [8]; according to the detection process. They can be divided into one-stage detection methods [9] and two-stage detection methods [10], [11], [12], according to the detection model with or without anchor, they can be divided into anchor-free detection methods [13] and anchor-based detection method [14]. Although the models with massive parameters proposed by those researchers have excellent object detection performance, they are all researched and developed on the ground server and do not take the extremely limited resources of the satellite platform into account, so those models cannot run normally on the satellite platform. Thus, how to design a lightweight detection architecture for an extremely resource-constrained satellite platform is the first key problem that needs to be addressed.

The second challenge is that the general lightweight object detection models had not been optimized according to the unique characteristics of RSIs that require specific network design. The generic lightweight object detection model is usually designed for natural images, which characterizes by high color saturation and rich object information. The RSI resolution changes unevenly or some pixels are missing caused of the oblique imaging view and various disturbances, leading the RSI defective and distorted. And RSIs have the characteristics of large scenes, God's perspective, and cross-scale objects, which also makes the object features of RSIs more difficult to extract than natural images. Therefore, general lightweight detection methods need to be adapted before they can be applied to RSIs. In addition, satellite on-orbit computing has the natural advantage of rapid response in urgent emergencies such as emergency rescue. This advantage is reflected not only in the lightness of the object detection model, but also in the detection efficiency and inference accuracy of the model. Therefore, the design of the network for on-orbit tasks needs to consider the tradeoff between efficiency and accuracy, and not just focus on the lightness of the network. However, existing lightweight models focus only on the lightweight of the model to accommodate edge devices with limited computational resources, ignoring the problem that in comparison with natural image features RSI features are difficult to extract. Therefore, how to design a lightweight and efficient object detection model for RSIs that adapt to the extremely resource-constrained satellite platform, while guaranteeing the feature extraction efficiency and inference accuracy, is another problem that needs to be addressed.

Based on the above challenges, we propose a lightweight accurate-oriented object detection network called SOCDet. The proposed scheme shows the characteristic of lightweight on-orbit detection, efficient feature extraction, and accurate inference. Specifically, our main contributions are summarized as follows.

1) We propose a lightweight and concise SOCDet method for satellite on-orbit computing, which satisfies the on-orbit environment of the extremely constrained satellite resources and achieves fast on-orbit inference and high-accuracy object detection.
2) We set up a structural reparameterization block (SRB) based on combined structure reparameterization to improve the inference speed without reducing the detection accuracy. We design Single-kernel Omni-dimensional Dynamic Convolution (SODC) and a plug-and-play efficient guided attention module (GAM) to improve the model object detection efficiency and model accuracy.
3) We conduct ablation experiments on the ground server and compare experiments with the state-of-the-art methods on the NVIDIA Jetson TX2 (TX2), using DOTA, HRSC2016, and FAIR1M. The experimental results show that SOCDet can achieve 2.36× faster than the baseline in inference speed with 10.78 more mAP, 72.6% fewer Params, and 92.56% fewer floating-point operations (FLOPs). Compared with state-of-the-art methods, SOCDet improves the inference speed and derives competitive mAP results with affordable computing capability.

The rest of this article is organized as follows. Section II introduces the related work to the SOCDet. Section III details our proposed lightweight accurate satellite-on-orbit object detection network, SOCDet. Section IV demonstrates the experimental results on the ground server and the TX2 satellite board to evaluate the effectiveness of the proposed approach. Finally, Section V draws conclusions.

II. Related Work

The storage and computing resources on satellites are limited and precious, so the satellite on-orbit intelligent object detection model must be a lightweight neural network model to adapt to the resource-constrained on-orbit processing platform. Second, high-time-sensitive application scenarios, such as rescue and disaster relief, emergency rescue, and battlefield situational awareness, require a faster response speed of satellite on-orbit computing. Therefore, the satellite on-orbit object detection model should be accurate, efficient, and fast. So, in this section, we review the related work in three aspects: 1) lightweight object detection network; 2) fast and efficient object detection; and 3) satellite on-orbit object detection.

A. Lightweight Object Detection Network

The satellite platform is an extremely resource-constrained edge device, so only a lightweight object detection network

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can run on its platform [15]. Huang et al. [16] proposed General Gaussian Heatmap Label Assignment with low parameter tuning and time costs to improve detection performance, and [17] designed an effective lightweight oriented object detector LO-Det for RSIs. The current object detection models based on deep learning are developing more complex and accurate. It is generally believed that the larger the number of the model parameters is, the higher the model complexity and accuracy will be. Model pruning technology, model distillation technology, model quantization technology, and model architecture optimization technology are the four main methods for deep-learning model compression and acceleration [18]. Studies have shown that there is redundancy in complex deep-learning models [19]. Based on this insight, researchers have studied pruning techniques [20] to prune network layers or network parameters that make less contribution to the efficiency and lightness of models. Some scholars are studying model distillation technology [21], [22], [23], using a relatively complex and huge model as a teacher model to guide a lightweight student model and obtain a lightweight and efficient student model through the method of knowledge transfer. Some scholars focus on weight quantization technology [24], [25], [26] by quantizing high-precision models into low-precision models through simple scaling mapping to achieve model compression and acceleration. Some scholars are exploring architecture optimization techniques, such as Neural Architecture Search [27], which achieves the goal of lightweight networks by designing efficient and lightweight network architectures [28].

However, those methods of model compression acceleration have some drawbacks. The model compression acceleration methods of model pruning and model quantization will reduce the accuracy while accelerating the model lightly. Model distillation needs to train a relatively large teacher model first as a guide for the student model, which increases the training cost on the ground. Most existing network architecture optimization methods are the result of network architecture search using reinforcement learning, and their search consumes high computational costs. To solve the problems of accuracy degradation and huge training costs associated with these model compression and acceleration methods, we designed a lightweight and concise SOCDet.

B. Fast and Efficient Object Detection

Satellite on-orbit computing has the natural advantage of rapid response in urgent scenarios such as emergency rescue. This advantage is reflected not only in the lightness of the object detection model, but also in the detection efficiency and inference speed of the model. The accurate and efficient inferencing of the object detection model depends not only on the lightweight of the model, but also on the efficient feature extraction capability of the model. Many scholars have studied efficient feature extraction methods, including structural reparameterization [29], [30], attention mechanism [31], and dynamic convolution [32]. The structural reparameterization technology reduces the memory access cost (MAC) of the model and improves the model inference speed by reparameterizing the efficient and complex training structure of the network into a simple inference structure. The attention mechanism is developed from the human visual attention mechanism, which can make the model more efficient to focus on key object areas. Dynamic convolution aims to balance network performance and computing load. Conventional approaches to improving network performance (wider and deeper) tend to result in higher computational costs and lower efficiency. Therefore, compared with static convolution, dynamic convolution has a stronger feature expression ability.

Structural reparameterization techniques, attention mechanisms, and dynamic convolution are all effective means to improve the efficiency of network feature extraction, and if they are combined to use, the results would be better. To further improve the efficient feature extraction capability of the model, we comprehensively use structural reparameterization, attention mechanism, and dynamic convolution to design an accurate inference block SRB with guaranteed feature extraction capability, a GAM that guides the model to focus on key object feature channels and key object feature regions, and an efficient feature extraction computational unit SODC.

C. Satellite On-Orbit Object Detection

As the satellite resolution is increased continuously, the quality of RSIs for earth observation is also improved significantly; however, the data volume is increasing, reaching several hundred Gb/s, and the bandwidth of satellite–ground data transmission is within tens of Gb/s, resulting in the problem that the data cannot be downlinked for application. Satellite on-orbit object detection is an effective way to solve this problem, and some scholars have already done some research on it. Zhou et al. [33] proposed a convolutional neural network-based object detection algorithm for RSIs, the computational cost of which is optimized to 25 M FLOPs by using transfer learning, sparse training, and weight quantification, while the weight file data are reduced from tens of Mb to 0.5 Mb. Wu et al. [34] used yolov5 for intelligent satellite on-orbit object detection with satellite–earth collaboration. Shi et al. [35] proposed an on-orbit ship (SH) detection and classification method for the SAR satellite, which adopted an adaptive sliding window to extract the domain connected to water and proposed the suspected target areas of the SH. Shen et al. [36] proposed fast and accurate multiclass geospatial object detection with large-size remote-sensing imagery using CNNs and truncated NMS. The research on satellite on-orbit object detection methods has focused on the lightweight of the model, but not enough attention has been paid to the computational efficiency of the model.

In this article, we propose SOCDet, a method that focuses not only on the lightweight of the model, but also on the efficiency, rapidity, and potential feature extraction capability of the model.

III. Methodology

SOCDet is a lightweight, accurate, and efficient object detection model that adopts a one-stage lightweight and concise detection architecture. The excellent feature extraction

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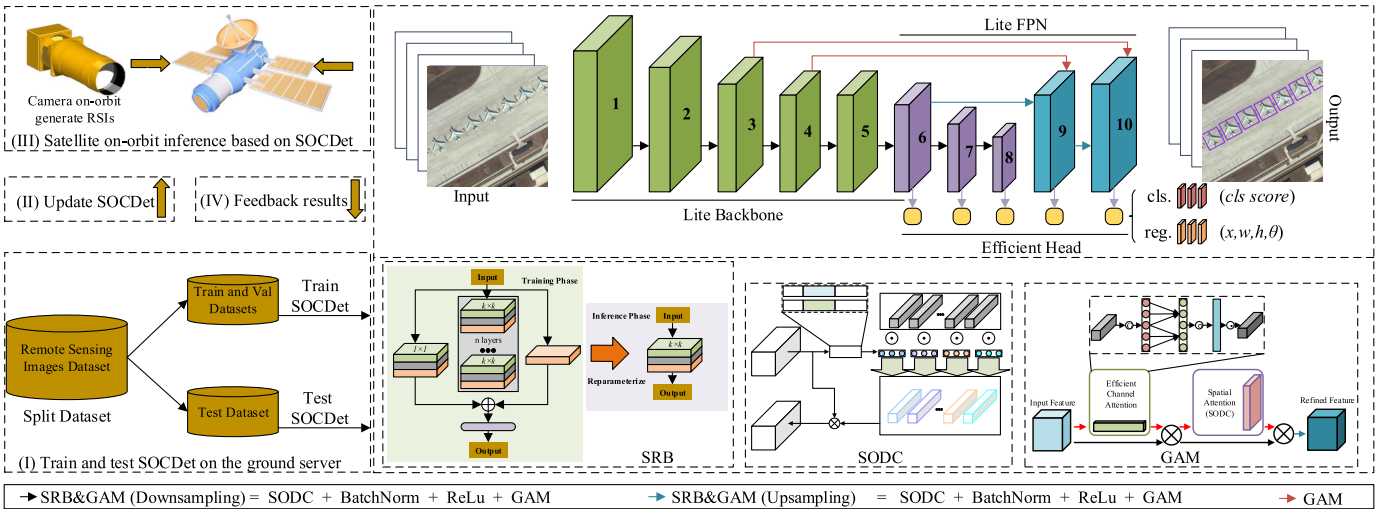


Fig. 1. Main flowchart of the proposed SOCDet. (I) Train and test SOCDet on the ground server. (II) Update SOCDet into the satellite. (III) Satellite on-orbit inference based on SOCDet. (IV) Feedback detection results to users and the ground server center. SOCDet consists of a lite backbone, lite FPN, and efficient head.

ability of SOCDet depends on the SRB, which is designed based on structural reparameterization technology to improve the detection speed and detection accuracy of the model. A GAM is embedded between the SRB layers, which can help the model efficiently extract key features that characterize object information. SODC is designed as the basic computing unit of SOCDet. Compared with other types of dynamic convolution, this convolution is lighter and more efficient.

In this section, we will describe in detail the lightweight concise detection architecture of SOCDet in Section III-A. The design principles and ideas of SRB, GAM, and SODC are explained in Sections III-B–III-D, respectively. Finally, we show details of the processing flow of satellite on-orbit oriented object detection based on SOCDet. The main flowchart of the proposed SOCDet is shown in Fig. 1.

A. Lightweight Concise Detection Architecture

To adapt the model to the extremely resource-constrained environment on the satellite, based on our evaluations of different architecture design choices, we developed a more concise and efficient one-stage oriented object detection architecture, as shown in Fig. 2. As the branch network structure needs a lot of memory access and will greatly affect the network inference speed, the Lite Backbone feature extraction network is designed as a stack of six-layer SRB and GAM combinations according to the single-branch simple stacking concept. And in this network, five downsampling operations are performed in Stages 1–4 and Stage 6. Stage 1 is obtained by applying a layer of Cov2D operation with kernel size = 3, filter = 64, stride = 1, and padding = 1, to the original input image. The specific parameters are shown in Table I. The Lite feature pyramid network (FPN) draws a branch from Stage 3, and Stage 4 of the Lite Backbone. The input of Lite FPN Stage 9 is the output of Stage 4 after the GAM operation and the output of Stage 6 after the SRB operation, and the output of Lite FPN Stage 9 is used as the input of Lite FPN Stage 10 through the SRB upsampling

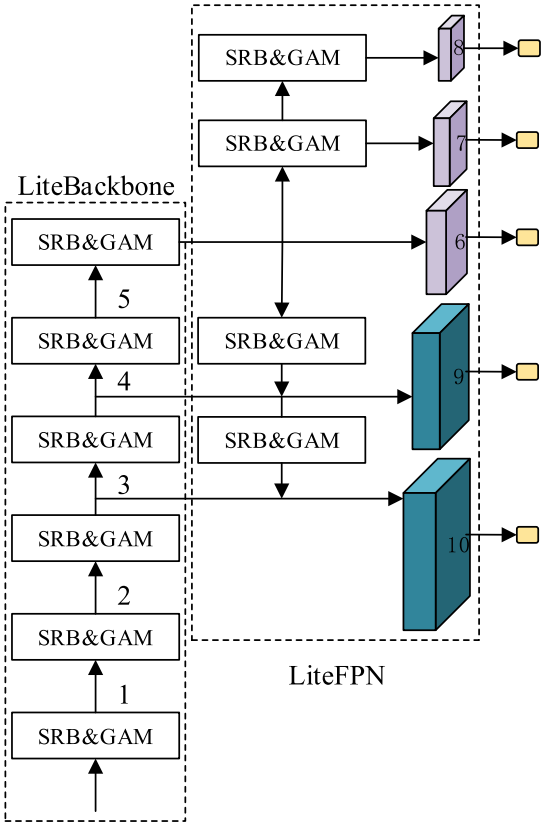


Fig. 2. SOCDet network structure diagram, as a supplementary description of Fig. 1 and consistent with Fig. 1. The arrow represents the convolutional features, and the rectangular box represents the operation.

operation, as shown in Fig. 2. The Efficient Head has a total of five detectors that can detect objects of various sizes. Each detector consists of an object classification subnetwork and a bounding box regression subnetwork. The object classification subnetwork and the bounding box regression subnetwork are both three-layer stacks of SRB and GAM. The difference is that the object classification subnetwork obtains the category of the object, and the bounding box regression subnetwork

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TABLE I
SOCDET BACKBONE NETWORK PARAMETER CONFIGURATION

Stage	Input	Blocks	Stride	Block Type	Channels
1	1024 × 1024	1	2	SRB	64
1	512 × 512	-	-	GAM	-
2	512 × 512	2	2	SRB	64
2	256 × 256	-	-	GAM	-
3	256 × 256	8	2	SRB	128
3	128 × 128	-	-	GAM	-
4	128 × 128	5	2	SRB	256
4	64 × 64	-	-	GAM	-
5	64 × 64	5	1	SRB	256
5	64 × 64	-	-	GAM	-
6	64 × 64	1	2	SRB	512
6	32 × 32	-	-	GAM	-

obtains the coordinates of the bounding box, and it can be expressed as $\{(x, w, h, \theta)\}$, where $x = (x_1, x_2)$ as the center of the bounding box, and w and h are the width and height of the bounding box, respectively. $\theta \in [-(\pi/4), (3\pi/4)]$ denotes the angle from the position direction of x_1 to the direction of the width w.

B. Composite Structure Reparameterization

To improve the inference speed of the model and ensure that the model has efficient feature extraction capabilities, we proposed composite structure reparameterization and designed an SRB based on it, which has a complex training structure and a simple inference structure. Both reducing model parameters and reducing model FLOPs can speed up model inference, but as a result, the inference accuracy is also reduced. Ding et al. [29], [30] believe that the main reason that affects the model inference speed is the MAC. The multibranch network structure can extract features more efficiently but require a high MAC. Structural reparameterization technology adopts reparameterizable multibranch operations in the training phase to improve feature extraction and reparameterizes the corresponding operations into simple operations without branches in the inferencing phase to reduce memory copies and lower MAC, improving model inferencing speed and accuracy without increasing FLOPs. The model complexity in the ground training phase increases, while the model complexity in the inference phase decreases.

Ding et al. [30] analyzed the homogeneity and additivity of the convolution and gives six types of transformations, and the ***Transform I (a conv for conv-BN)***, ***Transform II (a conv for average pooling*** and ***Transform IV (a conv for branch addition)*** of Fig. 3 are from [30]. We design ***Transform III: a conv for sequential $K \times K$ convolutions*** based on the study of [30].

Transform III: a conv for sequential convolutions. We can merge sequences of $K \times K$ conv, referring to any operation that can be transformed into a $K \times K$ convolution, with the same input and output dimension into one single $K \times K$ conv. We assume that the kernel shape of $K \times K$ conv-BN is $D \times D \times K \times K$, where D is the input and output dimension. We first fuse the two BN layers into the two conv layers to obtain $\boldsymbol{F}^{(1)} \in \mathbb{R}^{D\times D\times K\times K}$, $\boldsymbol{b}^{(1)} \in \mathbb{R}^{D}$, $\boldsymbol{F}^{(2)} \in \mathbb{R}^{D\times D\times K\times K}$, $\boldsymbol{b}^{(2)} \in \mathbb{R}^{D}$

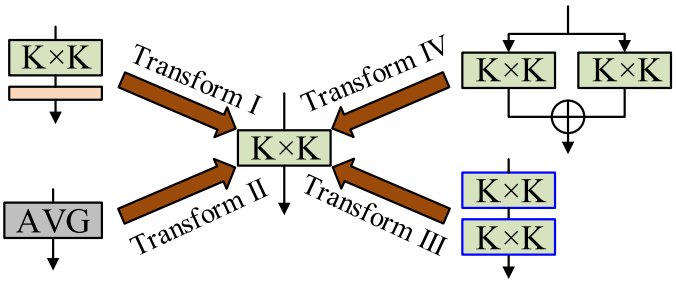


Fig. 3. Four transforms: different operations to convolutions. The green rectangle represents convolution, the gray rectangle represents average pooling, and the orange rectangle represents batch normalization.

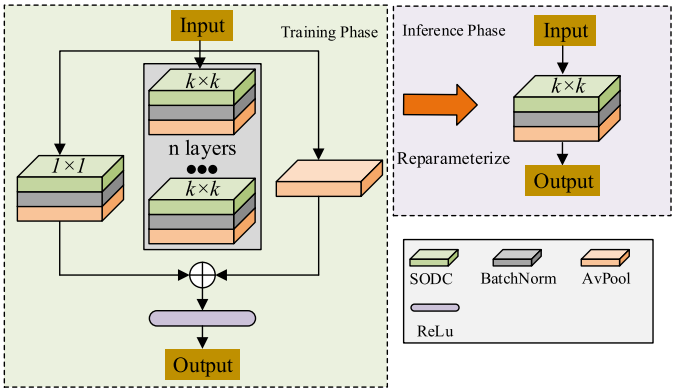


Fig. 4. Schematic of SRB. (Left) Training phase structure. (Right) Inference phase structure.

by using *Transform I*. The output of two $K \times K$ conv-BN is

$$\boldsymbol{O}' = \left(\boldsymbol{I} \circledast \boldsymbol{F}^{(1)} + \mathrm{REP}\left(\boldsymbol{b}^{(1)}\right)\right) \circledast \boldsymbol{F}^{(2)} + \mathrm{REP}\left(\boldsymbol{b}^{(2)}\right). \quad (1)$$

We expect that the expressions of the kernel and bias of a single conv, $\boldsymbol{F}'$ and $\boldsymbol{b}'$, can satisfy

$$\boldsymbol{O}' = \boldsymbol{I} \circledast \boldsymbol{F}' + \mathrm{REP}\left(\boldsymbol{b}'\right). \quad (2)$$

Applying the additivity of convolution to (1), we have

$$\boldsymbol{O}' = \boldsymbol{I} \circledast \boldsymbol{F}^{(1)} \circledast \boldsymbol{F}^{(2)} + \mathrm{REP}\left(\boldsymbol{b}^{(1)}\right) \circledast \boldsymbol{F}^{(2)} + \mathrm{REP}\left(\boldsymbol{b}^{(2)}\right). \quad (3)$$

We multiply the two convolution kernels $\boldsymbol{F}^{(1)} \circledast \boldsymbol{F}^{(2)}$ to obtain $\boldsymbol{F}'$

$$\boldsymbol{F}' \leftarrow \boldsymbol{F}^{(1)} \circledast \boldsymbol{F}^{(2)}. \quad (4)$$

The second term of (3) is convolution on constant matrices, so the outputs are also constant matrices. The third term of (3) is also a constant, so we can add the last two terms of (3) to get $\boldsymbol{b}'$

$$\mathrm{REP}\left(\boldsymbol{b}'\right) \leftarrow \mathrm{REP}\left(\boldsymbol{b}^{(1)}\right) \circledast \boldsymbol{F}^{(2)} + \mathrm{REP}\left(\boldsymbol{b}^{(2)}\right). \quad (5)$$

The design principle and formula derivation of SRB: SRB is implemented based on the composite structure reparameterization through the operations of reparameterization of the average pooling into convolution and a multibranch structure into a single-branch structure. SRB consists of two parts—training and inference—as shown in Fig. 4.

A three-branch structure is designed in the training part. Convolution-batch normalization-average pooling (Conv-BN-AVP) operations are common and efficient feature extraction

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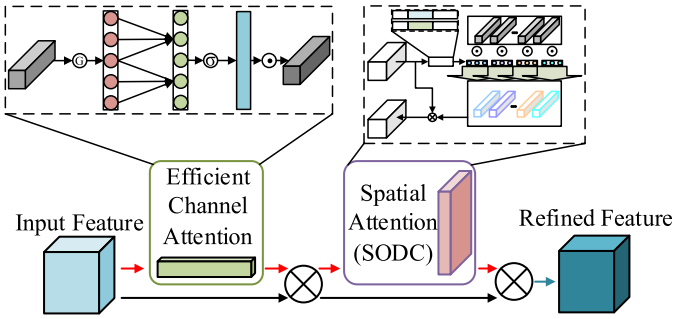


Fig. 5. Schematic of GAM.

operations in deep-learning models. The left branch adopts 1×1 Conv-BN-AVP operations to enhance the network's nonlinear feature extraction capabilities; the middle branch uses $k \times k$ Conv-BN-AVP stacking operations, which is the key to improving the feature extraction ability of the model, the number of stacked layers n is the hyperparameter of the network; the right branch adopts batch normalization operation to enhance the original feature expression; the left branch and right branch are similar to the shortcut operation in the ResNet, which avoids overfitting and gradient disappearance or gradient explosion to a certain extent.

The conv-BN of the left branch of Fig. 4 can be transformed into a 1×1 conv through *Transform I* and average pooling into a $k \times k$ conv through *Transform II*, and the 1×1 conv can be transformed to a $k \times k$ conv by means of padding [29], so those $k \times k$ convs can be transformed into one $k \times k$ conv through *Transform III*. The conv-BN-AVG of the middle branch of Fig. 4 can be transformed into a $k \times k$ conv through *Transform I–III*. The AvPool of the right branch of Fig. 4 can be transformed into a $k \times k$ conv through *Transform II*. The three $k \times k$ convs of the three branches of Fig. 4 can be transformed into a $k \times k$ conv through *Transform IV*.

The structure of the inference part is relatively simple, it is reparameterized by the structure of the training part into $k \times k$ Conv-BN-AVP operations. Because the convolution of 3×3 is very efficient, the convolution kernel of all Conv-BN-AVP operations in this article is $k = 3$. We analyze the effects of SOCDet in parameters, latency, speed, accuracy, and computing consumption before and after the SRB is used in Section IV-C5.

C. Guided Attention

To enable the model to extract the implied object information in RSIs accurately and efficiently, we innovatively designed a lightweight and efficient plug-and-play GAM based on the attention mechanism [31], as shown in Fig. 5. GAM adopts a lighter and more concise way to realize the channel attention mechanism to guide the model to pay more attention to the feature channels that are beneficial to the detection results and uses a more efficient SODC to guide the model to pay more attention to the area containing the object information when realizing spatial attention.

GAM uses a 1-D convolution operation to achieve efficient channel attention (ECA), allowing the model to focus more

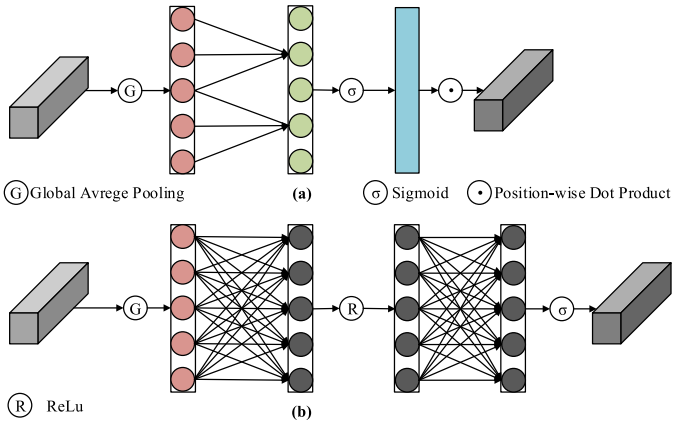


Fig. 6. Channel attention mechanism. (a) ECA. (b) SE.

on the focused object feature channel. The required number of parameters $\text{Conv1D}_{\text{params}}$ is derived as follows:

$$\text{Conv1D}_{\text{params}} = k_{\text{size}} \times C_{\text{in}} \times C_{\text{out}} \qquad C_{\text{in}} \leqslant C_{\text{out}}. \tag{6}$$

And the derivation of the number of parameters SE_{params} for the channel attention mechanism based on the fully connected layer is as follows.

$$\text{SE}_{\text{params}} = \frac{2}{r} \times C_{\text{out}}^2 \qquad 0 < r < 1. \tag{7}$$

Divide (6) by (7) to get

$$\text{Compression}_{\text{ratio}} = \frac{\text{Conv1D}_{\text{params}}}{\text{SE}_{\text{params}}} = \frac{r}{2} \times \frac{C_{\text{in}}}{C_{\text{out}}} < 1 \tag{8}$$

where C_{in} is the input feature channel and $k_{\text{size}} = 1$ is the kernel size of the 1-D convolution (when bias terms are used in Cov1D and SE, the introduced parameters and the computational cost is typically negligible). r is the reduction rate, and C_{out} is the number of output channels.

Fig. 6 shows the implementation of ECA and SE. The four operations required for the implementation of ECA are global average pooling, Conv1D, sigmoid, and position-wise dot product. The five operations required to implement SE are global average pooling, two fully connected layers, ReLu, and sigmoid. It can be seen that the implementation of ECA is simpler. Equation (6) shows that the number of parameters of the 1-D convolution is a first-order function of C_{out}, while (7) shows that the number of parameters of the fully connected layer is a second-order function of C_{out}. It can be seen from (8) that $\text{Compression}_{\text{ratio}}$ is less than 1, that is, as the number of output channels increases, the number of parameters of the fully connected layer channel attention will be much larger than that of the 1-D convolution channel attention. In addition, the 1-D convolution directly extracts the data features, unlike the fully-connected layer which requires feature scaling using the reduction rate r and then indirectly extracts the features, avoiding feature dimensionality reduction and efficiently extracting the features.

The efficient feature extraction computational unit SODC proposed in this article is used by GAM to implement spatial attention, and the detailed description can be found in Section III-D.

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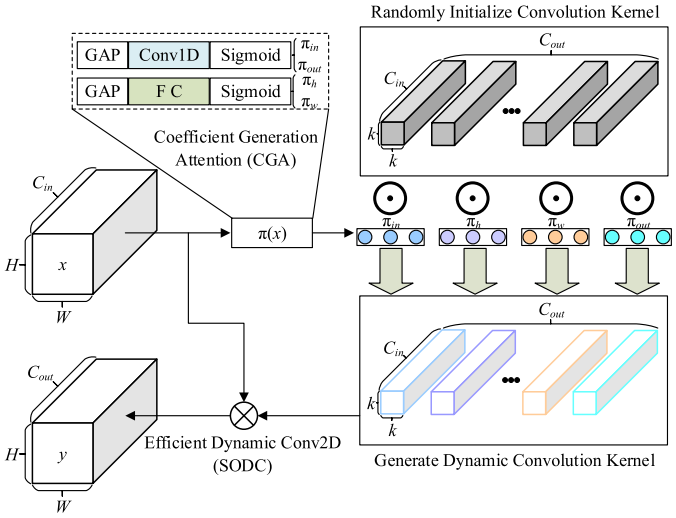


Fig. 7. Schematic of SODC.

D. Single-Kernel Omni-Dimensional Dynamic Convolution

The feature extraction capability of dynamic convolution is more efficient than that of standard convolution. To further improve the efficiency of the feature extraction ability of SOCDet, we designed a more efficient and lighter SODC.

Dynamic convolution can generally be understood as a linear combination of multiple randomly initialized convolution kernels, and the combination parameters are learned based on data samples. The specific derivation process of the dynamic convolution operation is as follows:

$$\mathrm{dc}(x) = (\alpha_1 W_1 + \cdots + \alpha_n W_n) \times x \tag{9}$$

where x represents the input features, α_n represents the nth weighting factor, and W_n represents the nth random initialized convolution kernel weight. The advantage of dynamic convolution over standard convolution is that dynamic convolution is more efficient in feature extraction. To fully exploit the efficiency and powerful feature extraction capability of dynamic convolution, we design an SODC, as shown in Fig. 7.

Different from the general dynamic convolution that linearly aggregates multiple initialized convolution kernels, SODC generates convolution 4-D scale coefficients $\Pi(\pi_{\mathrm{out}}, \pi_{\mathrm{in}}, \pi_h, \pi_w)$ of a randomly initialized convolution kernel $\mathrm{W} \in \mathbb{R}^{C_{\mathrm{out}} \times (C_{\mathrm{in}}/g) \times k \times k}$ through coefficient generation attention (CGA) learning input features, where $C_{\mathrm{out}}, (C_{\mathrm{in}}/g), k, k$ represent output channel, input_channel/groups, kennel height size, and kernel width size, respectively. CGA uses the Cov1D operation to generate the scale coefficient $\pi_{\mathrm{out}}, \pi_{\mathrm{in}}$, which is lighter and faster and uses the full connection operation to generate the scale coefficient π_h, π_w. The specific derivation process of SODC operation is as follows:

$$\Pi(x) = \begin{cases} \mathrm{Sigmoid}(\mathrm{Conv1D}(\mathrm{GAP}(x))), & x \in \mathbb{R}^{C_{\mathrm{in}}} \\ \mathrm{Sigmoid}(\mathrm{FC}(\mathrm{GAP}(x))), & x \in \mathbb{R}^{h \times w} \end{cases} \tag{10}$$

$$\mathrm{SODC}(x) = \Pi(x) \times W \times x + \pi_{\mathrm{out}} \times b \tag{11}$$

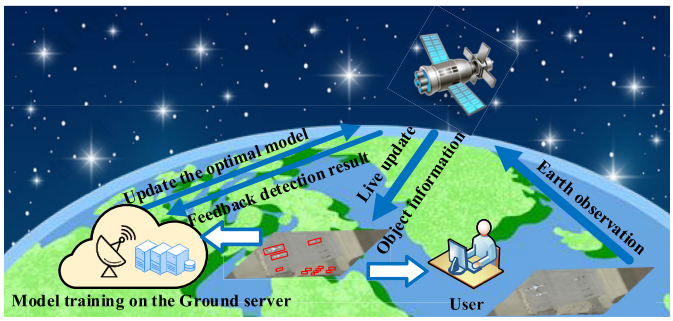


Fig. 8. Satellite on-orbit object detection based on SOCDet.

where $x \in \mathbb{R}^{C_{\mathrm{in}} \times h \times w}$ represents the input features, W is a randomly initialized convolution kernel weight, and b is a randomly initialized bias term.

E. Satellite On-Orbit Object Detection Based on SOCDet

The implementation process [3] of satellite on-orbit object detection technology is shown in Fig. 8, which is divided into four stages.

1) Training on the ground server. In the first stage, we construct the historical RSI dataset, use it to train and tune SOCDet on the ground server, and then obtain the optimal model to prepare for the second stage.
2) Updating the model on the satellite. In the second stage, we regularly update the latest model trained on the ground to the satellite. So the onboard model can detect new objects as early as possible. First, encapsulate the latest optimal recognition model in the ground system and send it to the satellite service through the satellite–ground uplink channel. After the satellite service system recognizes the data as the intelligent on-orbit processor's on-orbit data according to the agreed identification, the data is transmitted to the control unit of the intelligent on-orbit processing controller, and the control unit judges the validity of the received data, if it is valid, it will be transmitted to the computing unit, otherwise, it will be discarded and fed back by telemetry. Finally, the computing unit centrally caches the received data and performs model loading of the received data and on-orbit inferencing according to the control instruction after receiving is completed.
3) Inferencing on the satellite. In the third stage, we use the optimal model obtained in stage two and deploy it on the satellite board. The RSIs obtained on-orbit will be directly used for object detection without being transmitted to the ground.
4) Update object information live to users and service stations on the ground. In this stage, the satellite will send the detected object information, including object category, object coordinates, and object image slices to the users and the ground service center in real time.

This method of satellite on-orbit object detection has changed the traditional RSIs satellite-to-ground transmission and ground processing application mode, greatly improving the timeliness of satellite response.

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IV. Experimental Settings

In this section, the proposed SOCDet is experimentally verified on the ground server and the satellite imitated TX2 using three public datasets DOTA [37], HRSC2016 [38] and FAIR1M [39], respectively. The results of the experiments, including the ablation study and some comparison experiments of state-of-the-art lightweight methods, will be displayed and analyzed.

A. Data Description

1) DOTA [37]: It is a large RSI dataset for oriented object detection which contains 2806 RSIs with a size ranging from 800 × 800 pixels to 4000 × 4000 pixels, and 188 282 instances of 15 common object categories including Plane (PL), Baseball diamond (BD), Bridge (BR), Ground track field (GTF), Small vehicle (SV), Large vehicle (LV), SH, Tennis court (TC), Basketball court (BC), Storage tank (ST), Soccer-ball field (SBF), Roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC).

Both the training and validation sets are used for training, and the test set is used for testing. Following [37], we crop a series of 1024 × 1024 pixels patches from the original image with an overlap of 200.

2) HRSC2016 [38]: The HRSC2016 dataset comes from Google Maps, with a total of 1061 images. The training set, validation set, and test set contain 436, 181, and 444 images, respectively, with a total of 2976 objects, and the image resolution is between 2 and 0.4 m. Image sizes range from 300 × 300 pixels to 1500 × 900 pixels and most are larger than 1000 × 600 pixels. In the experiments, we resize all images to 800 × 800 pixels.

3) FAIR1M [39]: FAIR1M is a large-scale dataset for fine-grained object detection and recognition in RSIs. To meet the needs of practical applications, images in the FAIR1M dataset are collected from different sensors and platforms, with a spatial resolution ranging from 0.3 to 0.8 m. There are more than 1 million instances and more than 40 000 images in this dataset. All objects in the FAIR1M dataset are annotated for five categories and 37 subcategories by OBB. The size of images ranges from 1000 × 1000 pixels to 10 000 × 10 000 pixels, and each image contains objects exhibiting a wide variety of scales, orientations, and shapes. In this article, considering the limited processing capability of satellite, we crop the images into 1024 × 1024 pixels patches with a stride of 200.

B. Experimental Details

All experimental model training is carried out on the ground server. The server configuration is as follows: 2 × 10-core Intel Xeon[1] E5-2640v4 at 2.40 GHz CPU, an NVIDIA Corporation TU102 (GeForce RTX 2080 Ti) GPU, and an operating system of 64-bit Ubuntu 20.04.1 LTS, the programming language is Python-3.7.4, and the deep-learning framework is Pytorch-1.9.0.

[1]Registered trademark.

TABLE II
Results of Lightweight Concise Detection Architecture Ablation Experiments

Model	SRB	Attention	Conv	FPS	mAP	FLOPs(G)	Params(M)
RetinaNet [41]	-	-	conv	11.27	68.05	215.92	36.42
S2ANet [42]	-	-	conv	15.24	73.04	159.27	33.25
R3Det [9]	-	-	conv	12.41	69.81	198.81	34.76
QueryDet [43]	-	-	conv	13.15	69.26	173.25	33.91
SOCDet_A	-	-	conv	20.88	65.42	10.26	7.53

The training schedule is the same as that for mmdetection [40]. All models are trained in 12 epochs for DOTA and 36 epochs for HRSC2016. The SGD optimizer is adopted with an initial learning rate of 0.0025 and the learning rate is divided by 10 at each decay step. The momentum and weight decay are 0.9 and 0.0001, respectively. We adopt a learning rate linear warm-up for 500 iterations. The approximate SkewIoU loss [9] is adopted as a loss function, the Focal Loss hyperparameter $\gamma = 2.0, \alpha = 0.25$, and loss_weight = 1.0.

The experimental performance metrics are mean Average Precision (mAP), inference speed frames/s, the amount of computation (FLOPs), and the number of parameters.

The configuration of the TX2 development environment is as follows: The operating system is 64-bit Ubuntu 18.04.5 LTS, CUDA Version 10.2.89, the Jetpack version is Jetpack 5.1, the programming language is Python-3.6.9, and the deep-learning framework is torch-1.8.0 and torchvision-0.9.0a0.

C. Ablation Study

Aiming to highlight the contributions of the SOCDet proposed in this work in the lightweight, concise architecture SRB, GAM, and SODC as well as the benefits of such a design in the lightweight concise architecture, we set up related ablation experiments on a ground server.

1) Study of Lightweight Concise Detection Architecture: The method proposed in this article is a variant of RetinaNet [41], so when setting up lightweight concise detection architecture ablation experiments, RetinaNet (Baseline method) and the three latest pieces of research of RetinaNet [41], S2ANet [42], R3Det [9], and QueryDet [43] are selected as comparative experiments. To highlight the advantages of SOCDet's lightweight concise architecture as much as possible, SOCDet_A just has the same architecture as SOCDet and does not use SRB, GAM, and SODC proposed in this article.

As seen in Table II and Fig. 9, SOCDet_A reduces parameters by 79.32% over the RetinaNet baseline with a less significant reduction in mAP, reduces FLOPs by 95.24%, improves the inference speed by 85.27%, and achieves 20.88 frames/s on 1024 × 1024 images. SOCDet_A achieves better results than that of the other three variants of RetinaNet, S2ANet, R3Det, and QueryDet, in Prams, frames/s, and FLOPs. It can be proved that the architecture of SOCDet is lightweight and concise.

2) Study of Composite Structure Reparameterization: To highlight the contribution of SRB in model inference speed, we set up SOCDet_SRB without SRB for experimental

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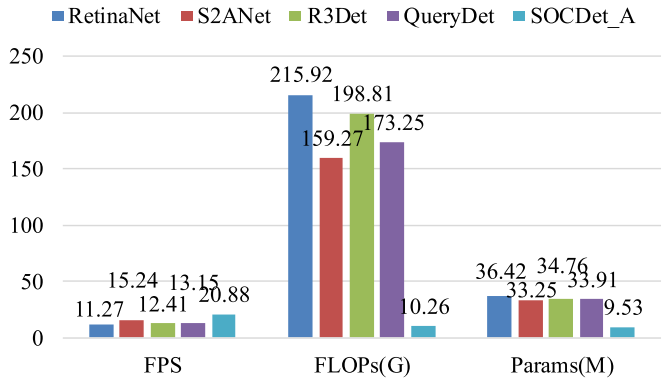


Fig. 9. Comparison of experimental results of the model structure.

TABLE III
RESULTS OF COMPOSITE STRUCTURE REPARAMETERIZATION ABLATION EXPERIMENTS

Model	SRB	Attention	Conv	FPS	mAP	FLOPs(G)	Params(M)
SOCDet_SRB	-	GAM	SODC	19.98	73.32	13.56	7.89
SOCDet	✓	GAM	SODC	**26.57**	**78.83**	16.06	9.98

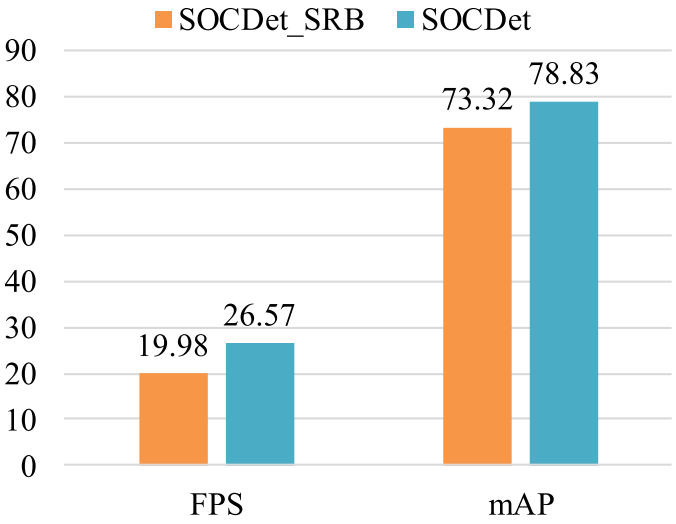


Fig. 10. Comparison of SRB ablation results.

comparison, while other settings are the same as in SOCDet. The experimental results are shown in Table III and Fig. 10.

From the experimental results in Fig. 10, we can find that the inference speed of SOCDet_SRB without SRB is reduced by 21.86%, and the mAP is reduced by 5.51 compared with SOCDet. From the experimental results in Table III, we can find that the inference speed of SOCDet_SRB without SRB is lower than that of the model with SRB. It can be proved that SRB can improve the inference speed and the mAP of the model to a certain extent.

3) Study of Guided Attention: To highlight the contribution of GAM to improve model mAP, we set up the comparative experiments SOCDet_CBAM, SOCDet_SE, and SOCDet_Att using CBAM [44], SE [45], and without the attention module, and other experimental conditions are the same as SOCDet.

From the attention plots in Fig. 11, we can find that the features extracted by SOCDet using GAM are mainly focused on the object, while the features extracted by SOCDet_CBAM, SOCDet_SE and SOCDet_Att do not match with the object exactly. From the experimental results in Table IV and Fig. 12, it can be seen that the accuracy of the model using GAM is higher than that of the model using the attention mechanism, and the SOCDet using GAM has the highest mAP, reaching 78.83. This proves that GAM improves the mAP of the model by guiding the model to pay more attention to key object regions and feature channels.

TABLE IV
RESULTS OF GUIDED ATTENTION ABLATION EXPERIMENTS

Model	SRB	Attention	Conv	FPS	mAP	FLOPs(G)	Params(M)
SOCDet_CBAM	✓	CBAM	SODC	25.03	72.36	16.07	9.92
SOCDet_SE	✓	SE	SODC	26.52	70.84	15.87	9.58
SOCDet_Att	✓	-	SODC	26.71	69.98	15.53	9.56
SOCDet	✓	GAM	SODC	**26.57**	**78.83**	16.06	9.98

TABLE V
RESULTS OF SODC EXPERIMENTS

Model	SRB	Attention	Conv	FPS	mAP	FLOPs(G)	Params(M)
SOCDet_Dy	✓	GAM	Dy	25.55	75.65	16.12	10.03
SOCDet_Cond	✓	GAM	Cond	24.02	76.68	16.32	10.15
SOCDet_Conv	✓	GAM	conv	26.82	75.32	15.89	9.89
SOCDet	✓	GAM	SODC	**26.57**	**78.83**	16.06	9.98

4) Study of SODC: To demonstrate the efficient feature extraction capability of SODC, we set up SOCDet_Conv, SOCDet_Dy, and SOCDet_Cond using standard convolution, dynamic convolution [46], and cond dynamic convolution [32], respectively, and other experimental conditions are the same as SOCDet.

It can be seen from Table V and Fig. 13 that SOCDet with SODC achieves the highest mAP of 78.83, which is 3.51% higher than SOCDet_Conv, 3.18% higher than SOCDet_Dy, and 2.15% higher than SOCDet_Cond when their Prams, FLOPs, and frames/s are very close. This proves that SODC has made a great contribution to the efficient feature extraction of SOCDet.

5) Discussion of the Cost and Gain of Each Improvement Step: To further illustrate the subtle changes brought by each step of improvement to SOCDet, we discuss the impact of the lightweight framework, SODC, SRB, and GAM on SOCDet, respectively, using RetinaNet as a baseline.

Table VI shows the cost and gain of each improvement step. SOCDet_A only uses the SOCDet architecture, compared with the baseline, its latency is reduced by 46%, its inference speed is nearly doubled, the computational cost is reduced by 95.24%, and its parameters are reduced by 74%. This shows that the SOCDet architecture greatly reduces the computational cost of SOCDet and greatly improves its inference speed, which provides a guarantee for its stable operation on satellites. SOCDet_AS replaces the standard convolution in SOCDet_A with SODC. Compared with SOCDet_A, its latency is slightly increased, the inference speed is slightly slower, and the computational cost and parameters are slightly increased, but its mAP is improved by 7.19. This shows that SODC can significantly improve the detection accuracy of SOCDet. SOCDet_ASS adopts SRB based on SOCDet_AS. Compared with SOCDet_AS, its inference latency is reduced

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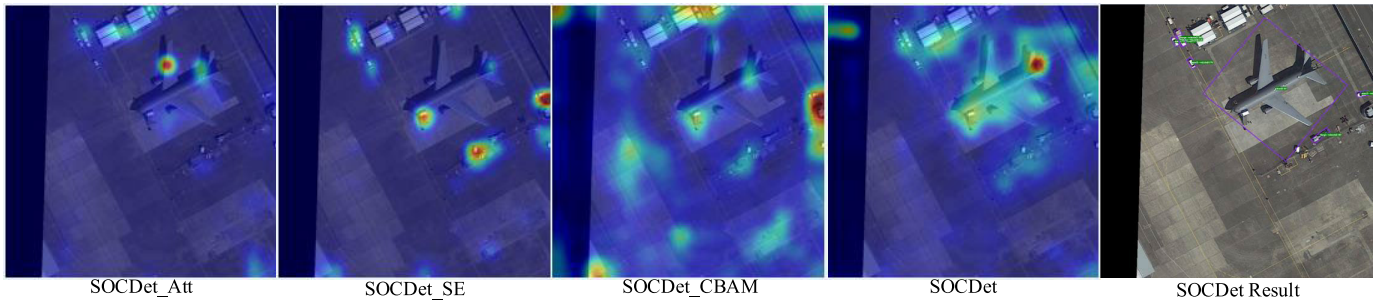


Fig. 11. Attention plots of different attention modules. SOCDet result represents the object detection result based on SOCDet.

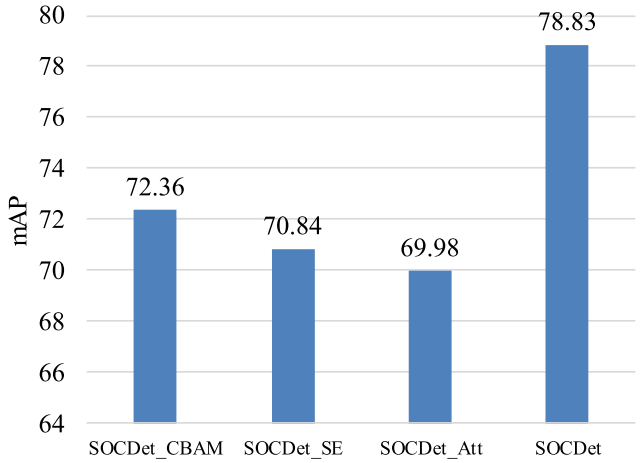


Fig. 12. Comparison of experimental results of different attention modules.

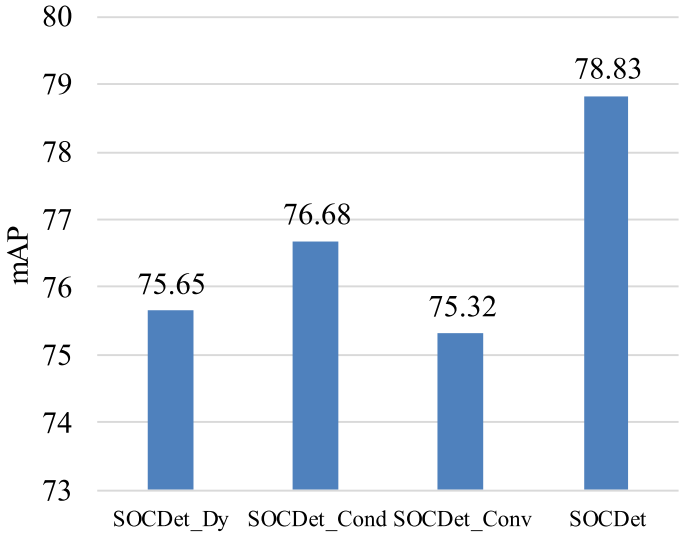


Fig. 13. Comparison of different convolution experimental results.

by 17.8%, its inference speed is increased by 21.68%, its mAP is increased by 1.57, and its computational cost and parameters are slightly increased. This shows that SRB can significantly improve the inference speed and detection accuracy of SOCDet. SOCDet introduces GAM based on SOCDet_ASS. Compared with SOCDet_ASS, its latency is reduced by 5.8%, its inference speed is improved, its mAP is increased by 4.65, and its computational cost and parameters are slightly increased. This shows that GAM can significantly improve the detection accuracy of SOCDet.

D. Comparison With State-of-the-Art Methods on the Satellite Imitated by TX2

To further prove the advantages of the SOCDet in efficiency and inference speed, we set up the existing advanced lightweight object detection networks YOLOv3tiny [47], YOLOv4tiny [48], YOLOv5s [49], YOLOXnano [50], NanoDet-M [51], NanoDet-plus-M [51], PP-PicoDet-S [52], YOLOv6nano [53], YOLOv7tiny [54], PP-YOLOEs [55], and oriented object detection methods specially designed for RSIs S2ANet [42], AOPG [56], and LO-Det [17] using DOTA, HRSC2016, and FAIR1M datasets on the satellite board TX2 for comparison experiments.

1) Results on DOTA: From the experimental results on the DOTA dataset in Table VIII and the detailed detection results of each category in Table VII, it can be seen that our proposed SOCDet improves the inference speed and derives the competitive mAP in the case of Params and FLOPs that TX2 can afford. The inference speed of SOCDet surpasses that of several other object detection models, including YOLOv3tiny (1.94× faster), YOLOv4tiny (1.82× faster), YOLOv5s (1.67× faster), YOLOXnano (1.71× faster), NanoDet-M (1.51× faster), NanoDet-plus-M (1.7× faster), PP-PicoDet-S (1.41× faster), YOLOv6nano (1.62× faster), and it is comparable to YOLOv7tiny in terms of inference speed. Additionally, SOCDet outperforms S2ANet (42× faster), AOPG (53.66× faster), and LO-Det (6.31× faster). Furthermore, the mAP of SOCDet is significantly higher than that of YOLOv3tiny (11.91 higher), YOLOv4tiny (8.36 higher), YOLOv5s (6.54 higher), YOLOXnano (12.97 higher), NanoDet-M (13.39 higher), NanoDet-plus-M (13.67 higher), PP-PicoDet-S (13.6 higher), YOLOv6nano (8.01 higher), and PP-YOLOEs (1.18 higher). SOCDet also outperforms S2ANet (5.79 higher), AOPG (3.44 higher), and LO-Det (3.59 higher). Despite requiring 16.06 FLOPs of computing resources and 9.98 M parameters, which are affordable for the TX2 system, SOCDet cannot outperform state-of-the-art models in all aspects. However, it demonstrates advantages in terms of inference speed and mAP. We can find that SOCDet has an excellent performance in densely arranged SH detection, near-shore SH detection with a complex background, and cross-scale detection of big aircraft and SVs from Fig. 14. The detailed detection results of each category are shown in Table VII.

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TABLE VI
COST AND GAIN OF EACH IMPROVEMENT STEP

Model	Architecture	SODC	SRB	GAM	Latency(ms)	FPS	mAP	FLOPs(G)	Params(M)
RetinaNet	-	-	-	-	88.73	11.27	68.05	215.92	36.42
SOCDet_A	✓	-	-	-	47.89(↓46%)	20.88(↑85%)	65.42(↓4%)	10.26(↓95%)	9.53(↓74%)
SOCDet_AS	✓	✓	-	-	48.61(↑1.5%)	20.57(↓1%)	72.61(↑11%)	11.28(↑10%)	9.79(↑3%)
SOCDet_ASS	✓	✓	✓	-	39.95(↓17.8%)	25.03(↑21%)	74.18(↑2%)	12.07(↑7%)	9.92(↑1%)
SOCDet	✓	✓	✓	✓	37.63(↓5.8%)	26.57(↑6%)	78.83(↑6%)	16.06(↑33%)	9.98(↑0.6%)

TABLE VII
DETAILED DETECTION RESULTS FOR EACH CLASS ON THE DOTA DATASET

Model	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
YOLOv3tiny	DarkNet	88.45	**85.07**	50.87	65.36	59.64	59.85	67.68	90.15	71.35	61.95	51.41	55.63	58.94	59.91	53.62	65.32
YOLOv4tiny	CSPDarkNet	89.04	81.36	53.67	**78.88**	61.36	71.28	75.36	89.06	72.54	72.78	54.64	57.84	57.82	62.22	55.26	68.87
YOLOv5s	CSPDarkNet	89.82	80.63	54.15	68.64	70.61	70.39	73.34	**90.86**	75.16	80.34	65.38	61.82	60.65	60.32	58.27	70.69
YOLOXnano	CSPDarkNet	86.87	79.64	52.27	65.98	57.89	61.91	59.32	88.69	70.69	61.48	52.06	55.28	56.03	61.14	54.71	64.26
NanoDet-M	ShuffleNetv2	86.42	78.56	51.18	64.25	56.38	59.03	68.32	89.62	71.67	59.62	49.65	54.85	57.21	58.38	52.59	63.84
NanoDet-plus-M	ShuffleNetv2	82.65	77.72	53.33	60.28	63.03	61.12	63.18	89.23	72.68	55.82	67.18	62.74	53.67	64.58	50.24	65.16
PP-PicoDet-S	LCNet	85.62	78.03	50.68	63.85	56.51	58.65	67.49	89.98	70.09	60.28	50.97	54.01	57.37	58.32	52.62	63.63
YOLOv6nano	EfficientRep	89.12	80.21	56.15	68.14	69.91	69.97	74.83	90.28	74.56	80.58	64.98	62.88	61.14	58.81	60.78	70.82
YOLOv7tiny	YOLOv7backbone	92.63	81.23	55.15	70.56	75.88	79.39	82.36	91.16	75.96	82.54	69.88	67.92	75.15	76.32	63.57	75.98
PP-YOLOEs	CSPRepResStage	89.87	82.36	57.29	75.18	79.09	80.07	83.82	88.79	81.96	85.48	72.03	70.28	77.23	77.12	64.22	77.65
S2ANet	ResNet50	89.57	80.58	56.62	69.17	73.34	76.62	75.68	89.62	75.67	80.82	69.62	62.27	70.32	65.35	60.39	73.04
APOG	ResNet50	89.62	79.93	57.18	74.95	75.68	79.86	78.67	89.64	79.45	82.98	70.16	61.81	72.57	75.72	62.68	75.39
LO-Det	MobileNetv2	89.22	66.14	31.32	55.96	70.05	71.04	84.27	90.74	75.09	81.28	44.65	59.34	59.98	65.13	48.42	66.17
SOCDet	SOCDetbackbone	**90.32**	84.64	**58.43**	78.62	**82.62**	**80.16**	**85.32**	90.58	**83.42**	**85.65**	**71.64**	**69.57**	**78.27**	**77.63**	**65.63**	**78.83**

TABLE VIII
RESULTS OF THE STATE-OF-THE-ART METHODS ON DOTA

Metric	YOLOv3tiny	YOLOv4tiny	YOLOv5s	YOLOXnano	NanoDet-M	NanoDet-plus-M	PP-PicoDet-S	YOLOv6nano	YOLOv7tiny	PP-YOLOEs	S2ANet	AOPG	LO-Det	SOCDet
Params(M)	8.68	5.96	7.06	**0.89**	0.91	1.07	1.05	4.1	6.01	7.72	33.01	33.85	6.75	9.98
FLOPs(G)	16.01	15.45	16.1	14.45	13.72	15.26	13.89	14.46	15.85	18.74	158.07	164.23	**10.96**	16.06
FPS	4.98	5.31	5.77	5.64	6.37	5.68	6.83	5.96	9.57	8.96	0.23	0.18	1.53	**9.66**
mAP	65.32	68.87	70.69	64.26	63.84	65.16	63.63	70.82	75.98	77.65	73.04	75.39	75.24	**78.83**

TABLE IX
RESULTS OF THE STATE-OF-THE-ART METHODS ON HRSC2016

Metric	YOLOv3tiny	YOLOv4tiny	YOLOv5s	YOLOXnano	NanoDet-M	NanoDet-plus-M	PP-PicoDet-S	YOLOv6nano	YOLOv7tiny	PP-YOLOEs	S2ANet	AOPG	LO-Det	SOCDet
Params(M)	8.71	5.99	7.09	**0.91**	0.93	1.10	1.08	4.12	6.04	7.75	33.05	33.87	6.78	10.01
FLOPs(G)	16.08	15.48	16.2	14.48	13.74	15.29	13.92	14.49	15.88	18.78	158.09	164.33	**10.98**	16.72
FPS	6.23	6.58	7.09	6.98	7.57	6.95	7.81	6.44	9.62	9.51	0.48	0.37	7.25	**10.82**
mAP	91.28	92.64	93.87	90.18	91.72	92.58	90.89	90.23	93.43	93.67	95.22	90.58	90.17	**94.28**

2) Results on HRSC2016: As can be seen from the experimental results on the HRSC2016 dataset in Table IX, SOCDet also improves the inference speed and derives the competitive mAP 94.28 on the HRSC2016 dataset under PASCAL VOC2012 metrics. The inference speed of SOCDet is 1.73× that of YOLOv3tiny, 1.64× that of YOLOv4tiny, 1.52× that of YOLOv5s, 1.55× that of YOLOXnano, 1.43× that of NanoDet-M, 1.55× that of NanoDet-plus-M, 1.39× that of PP-PicoDet-S, 1.68× that of YOLOv6nano, 1.12× that of YOLOv7tiny, 1.13× that of PP-YOLOEs, 22.54× that of S2ANet, 19.24× that of AOPG, and 1.49× that of LO-Det. The mAP of SOCDet is 3 higher than that of YOLOv3tiny, 1.64 higher than that of YOLOv4tiny, 0.41 higher than that of YOLOv5s, 4.1 higher than that of YOLOXnano, 2.56 higher than that of NanoDet-M, 2 higher than that of NanoDet-plus-M, 3.39 higher than that of PP-PicoDet-S, 4.05 higher than that of YOLOv6nano, 0.86 higher than that of YOLOv7tiny, 0.61 higher than that of PP-YOLOEs, 3.7 higher than that of AOPG, and 4.41 higher than that of LO-Det. Although the mAP of S2ANet is 0.94 higher than that of SOCDet, the inferencing speed of the former is only 4.43% of the latter. The visualization results of some state-of-the-art methods on HRSC2016 datasets are shown in Fig. 15. From Fig. 15, we can find that SOCDet has an excellent performance in near-shore SH detection with complex backgrounds.

3) Results on FAIR1M: FAIR1M is a fine-grained object detection dataset and contains many small objects with complex backgrounds and resolutions of only a few pixels, which brings great challenges to the object detection task. Table X shows the results of the state-of-the-art methods on FAIR1M. We can find that SOCDet improves the inference speed and derives the competitive mAP 34.53 on the FAIR1M dataset from Table X. The mAP of SOCDet is about 3 higher than that of YOLOv7tiny and PP-YOLOEs, 8.15 higher than that of NanoDet-plus-M, 0.91 higher than that of AOPG, and 4.25 higher than that of LO-Det. Although the mAP of AOPG is almost the same as that of SOCDet, and the mAP of S2ANet is even 0.07 higher than that of SOCDet, the inferencing speed of SOCDet is 46.85× that of AOPG and 42.59× that of S2ANet. Fig. 16 shows the visualization results of near-shore SH object detection. It can be seen from Fig. 16 that the characteristics of near-shore buildings and SHs are very similar

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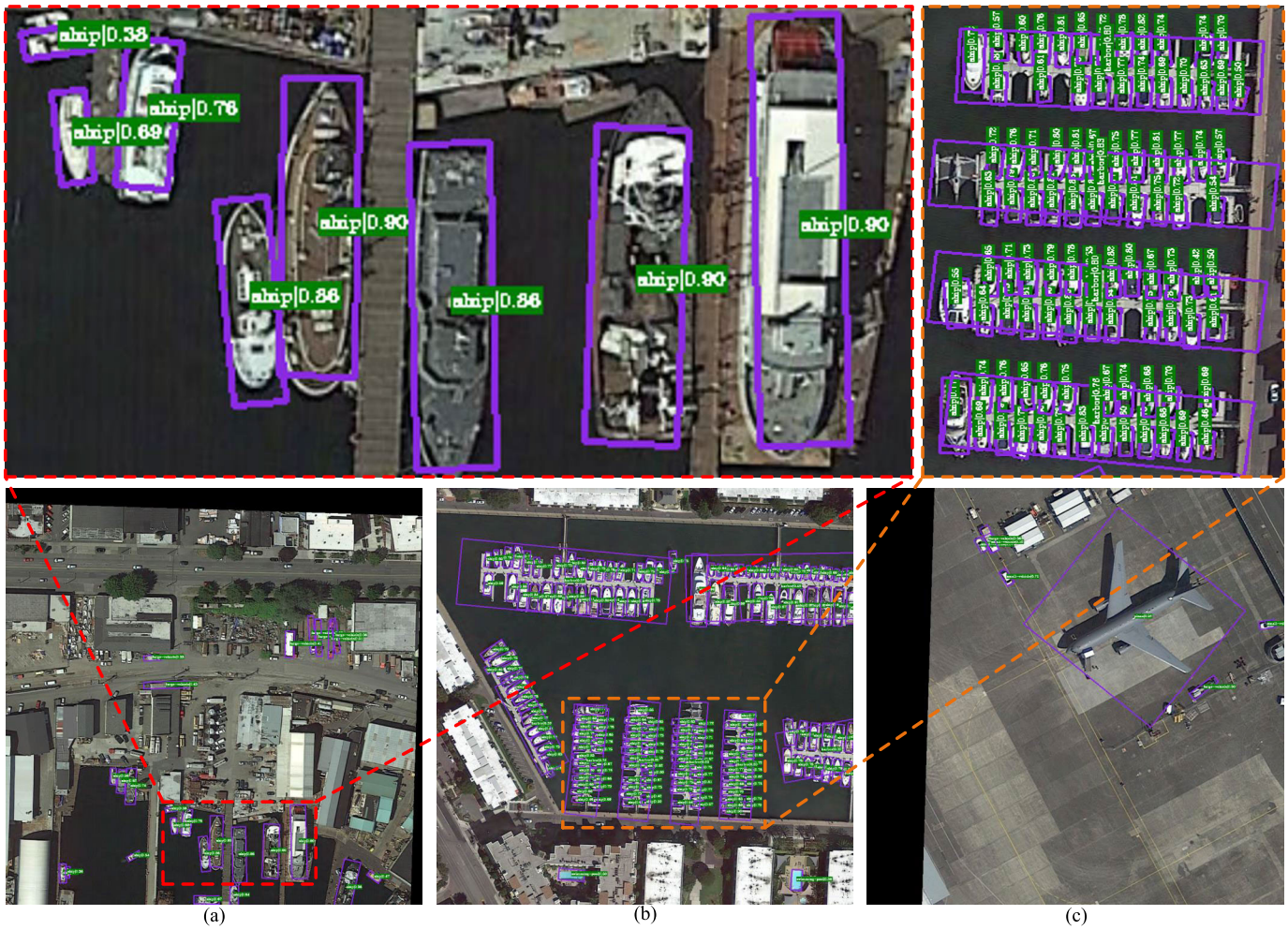


Fig. 14. Visualization results of SOCDet on the DOTA dataset. (a) Near-shore SH detection with a complex background. (b) Densely arranged SH detection. (c) Cross-scale detection of big aircraft and SVs.

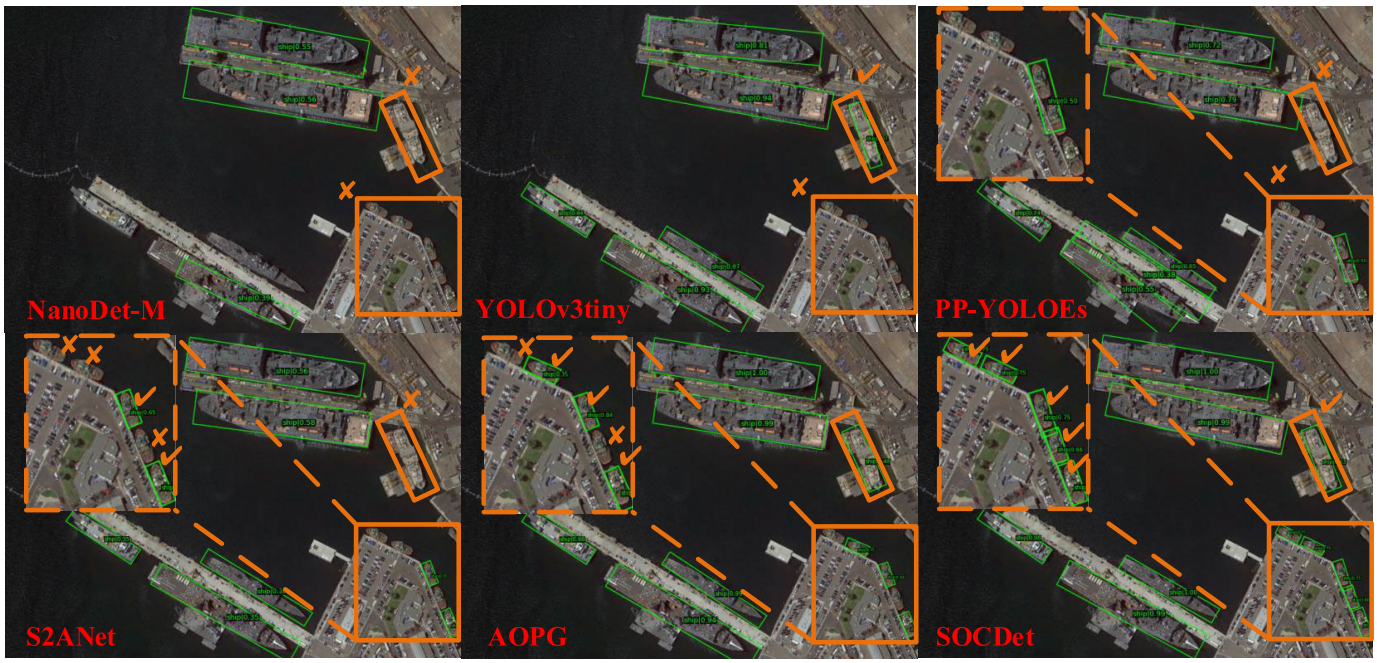


Fig. 15. Visualization results of some state-of-the-art methods on HRSC2016 dataset.

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TABLE X
RESULTS OF THE STATE-OF-THE-ART METHODS ON FAIR1M

Metric	YOLOv3tiny	YOLOv4tiny	YOLOv5s	YOLOXnano	NanoDet-M	NanoDet-plus-M	PP-PicoDet-S	YOLOv6nano	YOLOv7tiny	PP-YOLOEs	S2ANet	AOPG	LO-Det	SOCDet
Params(M)	8.86	6.06	7.13	**0.93**	0.95	1.17	1.18	4.3	6.2	7.93	33.25	34.07	6.93	11.28
FLOPs(G)	17.12	16.87	16.7	15.87	15.11	16.65	15.28	15.87	17.24	20.13	159.27	165.73	**12.36**	17.23
FPS	4.52	5.02	5.34	5.28	6.07	5.22	6.33	5.69	9.23	6.56	0.22	0.20	1.46	**9.37**
mAP	27.35	28.08	30.68	26.81	25.89	26.38	26.05	29.97	30.86	30.97	34.60	33.62	30.28	34.53



Fig. 16. Visualization results of the state-of-the-art methods on FAIR1M datasets.

in shape and color, which causes the AOPG method to directly miss the detection of SHs, while S2ANet mistakenly detects buildings as an SH, while SOCDet detects the SH accurately and has no false detections.

The performance of SOCDet on FAIR1M is not particularly excellent. The main reasons are: 1) FAIR1M has 37 fine-grained classifications, with small differences between different classes and great differences among the same classes, which leads to a certain degree of decline in detection accuracy and 2) each image of FAIR1M is large, the object on the image is very small, and the image background is complex. A single image even has 10 000 × 10 000 pixels, and the small object on the image has only a few pixels. Due to the interference information brought by the complex background, the detection accuracy is further reduced. Therefore, SOCDet still has the potential for better performance in fine-grained object detection and small object detection.

V. CONCLUSION

This article proposed SOCDet, a lightweight, efficient, and accurately oriented object detection network for satellite on-orbit computing. The SOCDet architecture adopts a lightweight and concise one-stage object detection approach by stacking the SRB and GAM combined operations. To improve the inference speed of SOCDet, the SRB is designed using structural reparameterization technology. This technology reduces the MAC during the inference phase, which significantly improves the inference speed without consuming additional FLOPs and Prams. To guide the model toward more efficient feature extraction, we designed a plug-and-play feature-GAM, GAM, which directs the model toward key object regions and feature channels. By doing so, the model achieves better feature extraction ability and improves the mAP. To further enhance the feature extraction efficiency, we designed an efficient convolutional computing unit, SODC, which achieves efficient feature extraction capability while increasing the consumption of almost negligible FLOPs. As a result, SODC brings an mAP gain of 3.51 to the model compared to standard convolution. The experimental results demonstrate that SOCDet achieves an inference speed 2.36× faster than the baseline with 10.78 more mAP, 72.6% fewer Params, and 92.56% fewer FLOPs. Compared to state-of-the-art methods, SOCDet improves the inference speed while also achieving competitive mAP results with affordable computing capability. Ablation experiments show the effectiveness and superiority of our proposed method on both global structure and local modules. Comparison experiments with state-of-the-art methods further demonstrate the advantages of our proposed method in inference speed and mAP.

In future research, we aim to enhance the SOCDet to not only detect objects on resource-constrained edge devices, but also achieve superior performance in fine-grained and small object detection. While the current version of SOCDet is suitable for resource-constrained edge devices, we intend to explore its potential in various other fields, such as drones, unmanned SHs, smart cameras, robots, autonomous driving, and so on. Besides improving the model structure, we also plan to investigate methods of model compression and acceleration to develop a lightweight model. Furthermore, we are steadfast in our pursuit of breakthroughs in brain-inspired computing and devoted to exploring the application of new brain-inspired computing technologies to edge devices.

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